

Unit 7 – Inference for Quantitative Data: Means

Notes 2: Significance Tests for Means

Important Ideas from Confidence Intervals for Means

- Proportions are used with _____ data, when you can split your data into “successes” and “failures”, and means are used with _____ data.
- When $n < 30$, and you check for “normality” by _____, you are not trying to prove that the sample data are normal, but rather trying to infer something about the population distribution.
- To use a t-distribution, we need the sampling distribution of $\bar{x}$ to be approximately normal, which is why we still check the “normal” condition despite using a t-statistic.
- You might see t^* written as _____, where the subscript represents the degrees of freedom.
- If your conditions are not met (not an SRS, 10% condition violated, data has outliers, etc.), DRAW ATTENTION TO IT, and continue with the inference procedure. In your conclusion, acknowledge that the conclusions might not be reliable because the conditions were not met.

Recall: Significance Tests

Hypotheses

- Null Hypothesis – A claim or a statement of “no difference” related to the population parameter
- Alternative Hypothesis – The situation we are trying to find evidence for, either greater than, less than, or different from the null hypothesis
- Example:
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Significance Level

- Our threshold for rejecting the null or failing to reject the null hypothesis
- $\alpha = 0.10, 0.05, \text{ or } 0.01$ usually

Test Statistic

- We find the test statistic by seeing how far away our sample statistic is from our null hypothesis

Test Statistic =

P-value

- Calculated from the test statistic, this tells us the probability we get this statistic (or more extreme in the direction of the alternative) if the null hypothesis is true.
- When _____, we can reject the null hypothesis in favor of the alternative hypothesis
- When _____, we fail to reject the null hypothesis and say we do not have evidence of the alternative hypothesis

$H_A:$	$H_A:$	$H_A:$

Conducting a Significance Test for a Mean

When conditions are met, the test statistic

for a _____ for a mean is:

- where t has _____ degrees of freedom
- $\bar{x}$ and s come from your sample data, and because of that, the formula $\frac{s}{\sqrt{n}}$ represents your standard error, and this is why a t-test statistic is needed
- if you know σ (which remember, you rarely do), we would perform a _____ for a mean:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

- Once we have our t-test statistic, we use tcdf(Lower Boundary, Upper Boundary, df) to find our p-value and then conclude in the context of the problem.

<i>Random</i>	<i>Independent</i>	<i>Normal</i>
OR	When sampling, check the 10% condition; In an experiment, we do not have to check this	_____, the _____ tells us it will be approximately normal _____, we have to either: 1) Have the original population be approx normally distributed 2) Graph the sample data to make sure there are no outliers or strong skewness

Example: One Sample T Test for a Mean

Your AP Statistics teacher believes that the average number of minutes their students spend on homework every night is greater than 35 minutes. She randomly selects a sample of 10 students and records the number of minutes they spent on homework last night. The data are below.

32	28	48	33	34	59	62	45	38	41
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a) Is there convincing evidence, at the 10% significance level, of your teacher's beliefs?

b) Describe a Type I and Type II error in the context of the problem above.